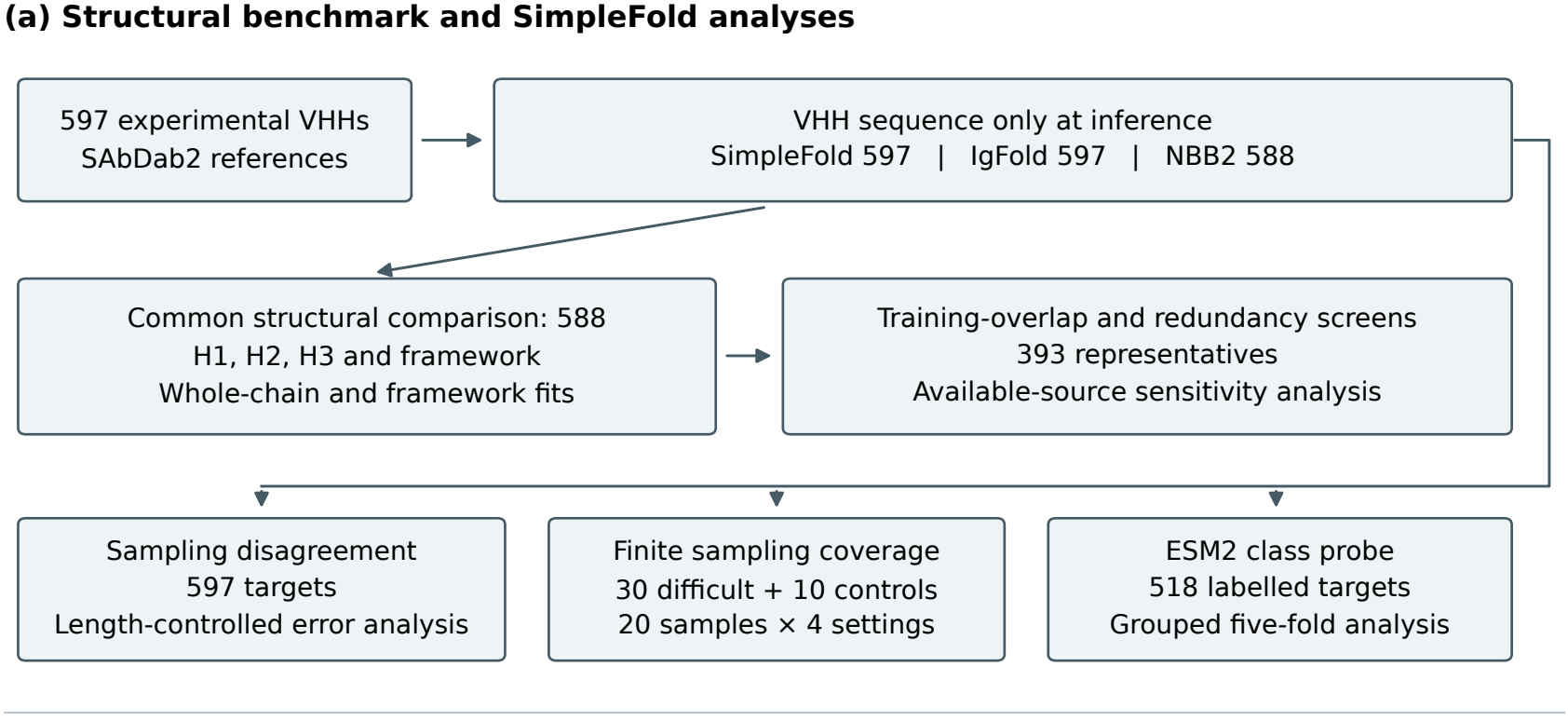
(a) Structural benchmark and SimpleFold analyses
597 experimental VHHs
SAbDab2 references
VHH sequence only at inference
SimpleFold 597 | IgFold 597 | NBB2 588
Common structural comparison: 588
H1, H2, H3 and framework
Whole-chain and framework fits
Training-overlap and redundancy screens
393 representatives
Available-source sensitivity analysis
Sampling disagreement
597 targets
Length-controlled error analysis
Finite sampling coverage
30 difficult + 10 controls
20 samples × 4 settings
ESM2 class probe
518 labelled targets
Grouped five-fold analysis
(b) Separate illustrative docking case: one complex
Experimental VHH control
and three predicted VHHs
from the 7R73 case
Fixed bound antigen
Same HDOCK protocol
Top 10 poses per input
DockQ and components
Epitope recovery
Descriptive case results